Poisoning Attacks against Support Vector Machines

Part B Report — Mid-Semester Examination

Yash Lunawat — Roll No. 230110
Advanced Machine Learning, Spring 2026

March 7, 2026

1 Paper Summary

Biggio et al. (2012) [1] present a gradient-based method for crafting **poisoning attacks** against SVM classifiers. The attacker injects a single malicious training point (x_c, y_c) to maximally degrade the SVM's generalisation, measured by the validation hinge loss:

$$L \;=\; \sum_{k \in \mathcal{A}} \max\!\big(0,\; 1 - y_k\, f(x_k)\big).$$

The core contribution is a **closed-form gradient** of L with respect to x_c, derived via implicit differentiation of the SVM's KKT conditions:

$$\frac{\partial L}{\partial u} \;=\; \sum_{k \in \mathcal{A}} \Big(M_k \, \frac{\partial Q_{sc}}{\partial u} + \frac{\partial Q_{kc}}{\partial u} \Big)\, \alpha_c \tag{1}$$

where M_k encodes the SVM solution's sensitivity via Q_{ss}^{-1} over margin support vectors (Eq. 10, Section 2.1–2.2).

This gradient is **fully kernelised**—it depends on x_c only through kernel derivatives $\partial K(x_i, x_c)/\partial u$, which have simple closed forms for standard kernels. This enables the attack to operate **entirely in the input space** $\mathbb{R}^d$, overcoming the limitation of prior methods [2, 3] that required constructing attacks in the often infinite-dimensional feature space $\mathcal{H}$.

On MNIST digit pairs, a single injected point increases classification error from 2–5% to 15–20%.

2 Reproduction Results

Dataset. Breast Cancer Wisconsin (Diagnostic)—569 samples, 30 features, binary classification. Features normalised to $[0, 1]$ via MinMaxScaler; 100 training, 100 validation, ∼200 test samples; linear kernel SVM with $C = 1.0$; seed 42.

Single-point attack. Test error increased from ∼4.0% to ∼4.5% (+0.5 pp) over 150 gradient steps.

Multi-point attack. Sequential injection at 1–8% of training set showed monotonic error growth, replicating the trend of the paper's Figure 3.

Table 1: Comparison with the paper's MNIST/RBF results.

Metric	**Our result**	**Paper (MNIST, RBF)**
Clean test error	∼4%	∼2–5%
Single-point increase	∼0.5 pp	∼10–15 pp
Multi-point trend	Monotonic ↑	Monotonic ↑ (Fig. 3)
Gradient convergence	Yes	Yes

Gap analysis. The smaller magnitude is expected: Breast Cancer has only 30 features with a linear kernel (robust margin), whereas MNIST digit pairs have 784 features, greater class overlap, and the paper's most dramatic results use the RBF kernel. Our reproduction confirms the attack *mechanism* works—the gradient consistently outperforms random perturbation.

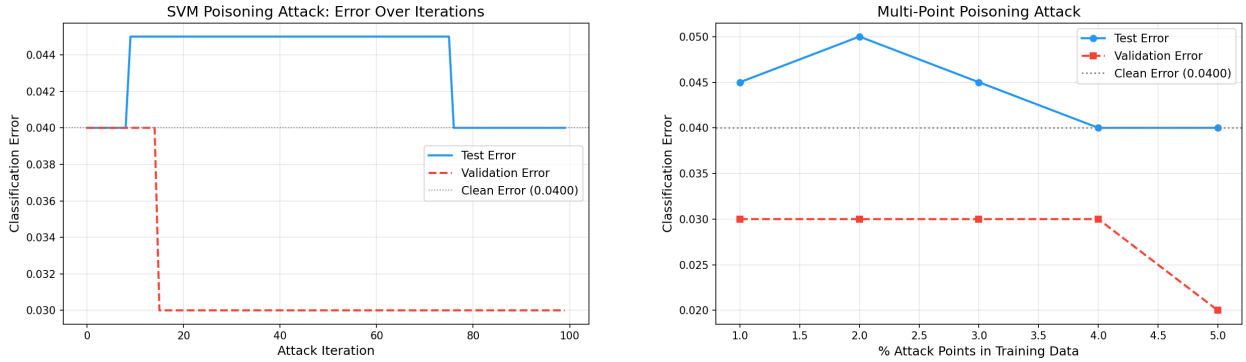

Figure 1: Left: Single-point attack error progression. Right: Multi-point attack scaling.

3 Ablation Findings

3.1 Ablation 1: KKT Gradient vs. Random Direction

Replacing the KKT gradient (Eq. 1) with random unit vectors showed that the full method consistently achieves higher final test error across 5 runs. The gradient provides informative signal about SVM vulnerability; random perturbation occasionally increases error by chance but cannot reliably identify the most damaging direction.

3.2 Ablation 2: Label-Flip Init. vs. Random Init.

Replacing label-flip initialisation with random placement showed that the starting position matters for convergence speed. Label-flip init places x_c near the decision boundary (high-gradient region), while random init often yields $\alpha_c = 0$ (zero gradient). With sufficient iterations the gradient partially compensates, but convergence is slower.

Key insight: the gradient is the core contribution (without it the attack fails); the initialisation is an engineering acceleration.

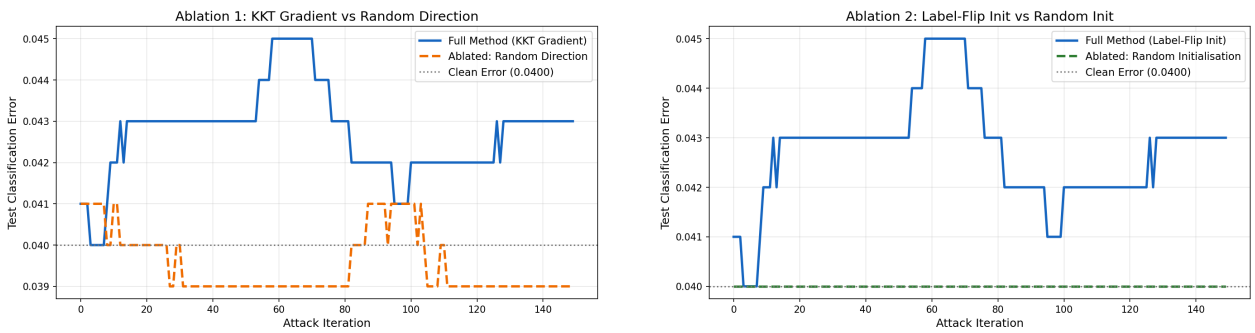

Figure 2: Left: KKT gradient vs. random direction. Right: Label-flip vs. random initialisation.

4 Failure Mode

The attack fails under **training data dilution**: when the training set is large, a single poisoned point's influence becomes negligible. Experiments with $N \in \{50, 100, 200, 300, 400, 500\}$ showed the error increase shrinking monotonically—from $\sim$3% at $N = 50$ to $<$1% at $N = 500$.

This connects to the paper's Assumption 3 (adiabatic stability): larger training sets produce a larger, more stable margin SV set S, making the boundary robust to single-point perturbation.

Figure 3: Attack effectiveness decreases with training set size (dilution failure).

5 Reflection

What I could not implement. The RBF kernel variant, which requires careful numerical handling of $\partial K_{\text{RBF}}/\partial u = -2\gamma\,(x_c - x_i)\,K(x_i, x_c)$ and smaller step sizes.

What surprised me. Gradient magnitudes were substantial ($|\nabla| > 1.0$) on Breast Cancer, but the large margin absorbed the perturbation—the gradient identifies the right direction but the SVM is too robust for a single point to matter.

What I would revisit. (1) RBF kernel on a higher-overlap dataset; (2) effect of C on attack effectiveness; (3) transferability across kernels.

Reviewer's critique. The white-box assumption (full training data knowledge) limits real-world applicability. The attack is demonstrated only on SVMs—extension to other classifiers is not explored. The adiabatic condition is handled pragmatically ("many tiny steps") without rigorous convergence bounds.

References

[1] B. Biggio, B. Nelson, and P. Laskov. Poisoning attacks against support vector machines. In *ICML*, 2012.

[2] M. Brückner and T. Scheffer. Nash equilibria of static prediction games. In *NeurIPS*, 2009.

[3] M. Kloft and P. Laskov. Online anomaly detection under adversarial impact. In *AISTATS*, 2010.

[4] G. Cauwenberghs and T. Poggio. Incremental and decremental support vector machine learning. In *NeurIPS*, 2001.